

Lecture 7: Normal Distribution, Joint PDFs

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1 Normal Distribution

1.1 Why the normal distribution?

Thus far, we've seen two continuous distributions – the **Uniform** (equally likely along an interval) and the **Exponential** (waiting times, memorylessness).

The **Normal** (or **Gaussian**) distribution is a little different than these – it arises naturally as the *universal limit* of averaging many small, independent effects — measurement error, heights, exam scores, noise in a communication channel. We'll make this precise later in the course (the **Central Limit Theorem**).

It is (one of) the most important distribution(s) in probability and statistics.

1.2 Normal random variables: definition

Definition: Normal random variable

A continuous random variable X is **normal** (or **Gaussian**) with parameters $\mu \in \mathbb{R}$ and $\sigma^2 > 0$, written $X \sim N(\mu, \sigma^2)$, if it has PDF

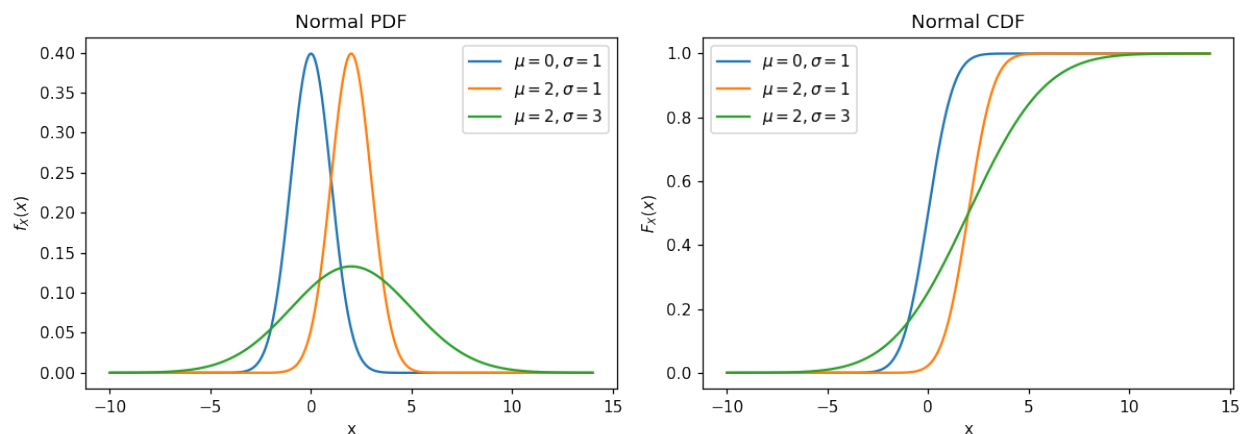
$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

- The density is a symmetric “bell curve” centered at μ .
- σ controls the spread: small σ gives a tall, narrow bump; large σ gives a short, wide one.
- It’s not obvious from the formula alone that this integrates to 1 — that relies on the classical **Gaussian integral** $\int_{-\infty}^{\infty} e^{-u^2/2} du = \sqrt{2\pi}$, which we’ll take as known and verify numerically below.

```
# Visualizing the PDF and CDF for a few choices of (mu, sigma)
xs = np.linspace(-10, 14, 500)
params = [(0, 1), (2, 1), (2, 3)]

fig, axes = plt.subplots(1, 2, figsize=(11, 4))
for mu, sigma in params:
    axes[0].plot(xs, stats.norm.pdf(xs, mu, sigma), label=f"$\mu={mu}, \sigma={sigma}$")
    axes[1].plot(xs, stats.norm.cdf(xs, mu, sigma), label=f"$\mu={mu}, \sigma={sigma}$")

axes[0].set_title("Normal PDF"); axes[0].set_xlabel("x"); axes[0].set_ylabel(r"$f_X(x)$"); axes[0].legend()
axes[1].set_title("Normal CDF"); axes[1].set_xlabel("x"); axes[1].set_ylabel(r"$F_X(x)$"); axes[1].legend()
plt.tight_layout()
plt.show()
```



1.2.1 Mean and variance of normal RV

Fact

If $X \sim N(\mu, \sigma^2)$, then

$$E[X] = \mu, \quad \text{Var}(X) = \sigma^2.$$

$E[X] = \mu$: substitute $u = x - \mu$. The density becomes symmetric in u about 0, i.e. $f_X(\mu + u) = f_X(\mu - u)$, and

$$E[X] - \mu = E[X - \mu] = \int_{-\infty}^{\infty} u \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-u^2/(2\sigma^2)} du = 0,$$

because the integrand is an **odd function** of u .

$\text{Var}(X) = \sigma^2$: this takes an integration-by-parts argument on $\int u^2 e^{-u^2/(2\sigma^2)} du$.

1.2.2 Normality is preserved by linear transformations

i Theorem

If $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$ for constants $a \neq 0, b$, then

$$Y \sim N(a\mu + b, a^2\sigma^2).$$

Proof

Recall if g is invertible and differentiable with inverse $h = g^{-1}$, then $f_Y(y) = f_X(h(y)) |h'(y)|$. Here $g(x) = ax + b$, so $h(y) = (y - b)/a$ and $h'(y) = 1/a$. Then

$$f_Y(y) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\left(\frac{y-b}{a} - \mu\right)^2}{2\sigma^2}\right) \cdot \frac{1}{|a|} = \frac{1}{|a|\sigma\sqrt{2\pi}} \exp\left(-\frac{(y - (a\mu + b))^2}{2a^2\sigma^2}\right),$$

which is exactly the $N(a\mu + b, a^2\sigma^2)$ density. ■

A linear function of a normal random variable is *again normal* — this is a special structural property that most distributions don't have (e.g., a linear function of a Uniform or Exponential RV is generally *not* Uniform or Exponential).

1.3 The standard normal random variable

i Definition: Standard normal

$Z \sim N(0, 1)$ is the **standard normal** random variable. Its PDF and CDF are denoted by lower- and uppercase phis:

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad \Phi(z) = \int_{-\infty}^z \varphi(t) dt.$$

Φ has no closed form in terms of elementary functions — historically this is why every probability textbook has a **standard normal table**:

z	0.00	0.50	1.00	1.50	1.645	2.00
$\Phi(z)$	0.5000	0.6915	0.8413	0.9332	0.9500	0.9772

You can use `scipy.stats.norm` to play the role of the table, but the *method* — standardize, then look up — is exactly what you'd do with a printed table.

1.3.1 Standardizing a normal random variable

i Corollary (of the linear-transformation theorem)

If $X \sim N(\mu, \sigma^2)$, then

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

This is immediate from the theorem two slides ago with $a = 1/\sigma$, $b = -\mu/\sigma$: $a\mu + b = 0$ and $a^2\sigma^2 = 1$.

This lets us answer probability questions about *any* normal $X \sim N(\mu, \sigma^2)$ using only the single function Φ !

$$P(X \leq x) = P\left(\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

1.3.2 Example: snowfall

Annual snowfall X (in inches) at a mountain resort is modeled as $X \sim N(60, 20^2)$.

(a) What's the probability of more than 90 inches of snow this year?

$$P(X > 90) = 1 - \Phi\left(\frac{90 - 60}{20}\right) = 1 - \Phi(1.5).$$

(b) How much snow is exceeded in only 5% of years — i.e. find x^* with $P(X > x^*) = 0.05$?

$$1 - \Phi\left(\frac{x^* - 60}{20}\right) = 0.05 \iff \Phi\left(\frac{x^* - 60}{20}\right) = 0.95 \iff \frac{x^* - 60}{20} = 1.645$$

using the table value $\Phi(1.645) \approx 0.95$ — so $x^* \approx 60 + 20(1.645) = 92.9$ inches.

```
mu, sigma = 60.0, 20.0

# (a)
p_over_90 = 1 - stats.norm.cdf((90 - mu) / sigma)
print(f"(a) P(X > 90) = {p_over_90:.4f}    [table method: 1 - Phi(1.5) = {1 - 0.9332:.4f}]")

# (b) 95th percentile
z95 = stats.norm.ppf(0.95)
x_star = mu + sigma * z95
print(f"(b) z with Phi(z)=0.95: {z95:.3f}")
print(f"    x* (95th percentile) = {x_star:.2f} inches")
```

(a) $P(X > 90) = 0.0668$ [table method: $1 - \Phi(1.5) = 0.0668$]

(b) z with $\Phi(z)=0.95$: 1.645

x^* (95th percentile) = 92.90 inches

1.3.3 Example: signal detection

A transmitter sends a bit $s \in \{-1, +1\}$ (equally likely) over a noisy channel. The receiver observes

$$Y = s + N, \quad N \sim N(0, \sigma^2) \text{ independent of } s,$$

and decides “+1 was sent” if $Y > 0$, and “−1 was sent” otherwise.

⚠ Question

What is the probability the receiver makes an error?

By symmetry it suffices to compute the error probability given $s = +1$ was sent (the case $s = -1$ is identical by symmetry of the noise):

$$P(\text{error} \mid s = 1) = P(Y < 0 \mid s = 1) = P(1 + N < 0) = P(N < -1) = \Phi\left(\frac{-1}{\sigma}\right).$$

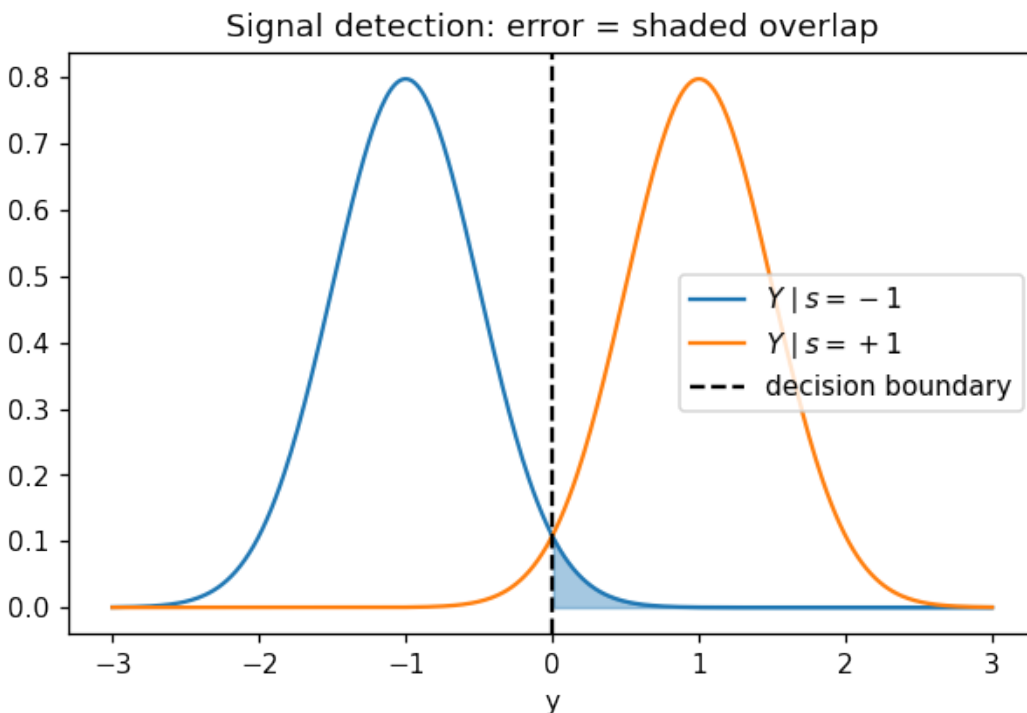
So $P(\text{error}) = \Phi(-1/\sigma)$ — smaller noise σ (relative to the signal strength 1) means a more reliable channel, exactly as intuition demands.

```
sigma = 0.5
p_error_formula = stats.norm.cdf(-1 / sigma)
print(f"P(error) = Phi(-1/sigma) = {p_error_formula:.4f}    (sigma = {sigma})")

# Verify by simulation
n = 2_000_000
s = rng.choice([-1, 1], size=n)
N = rng.normal(0, sigma, size=n)
Y = s + N
decision = np.where(Y > 0, 1, -1)
p_error_sim = np.mean(decision != s)
print(f"P(error), simulated      = {p_error_sim:.4f}")

# Visualize the two conditional densities and the decision boundary
y_grid = np.linspace(-3, 3, 400)
plt.figure(figsize=(6.5, 4))
plt.plot(y_grid, stats.norm.pdf(y_grid, -1, sigma), label=r"$Y \mid s=-1$")
plt.plot(y_grid, stats.norm.pdf(y_grid, 1, sigma), label=r"$Y \mid s=+1$")
plt.axvline(0, color="k", linestyle="--", label="decision boundary")
plt.fill_between(y_grid[y_grid > 0], 0, stats.norm.pdf(y_grid[y_grid > 0], -1, sigma),
                 alpha=0.4, color="C0")
plt.legend(); plt.xlabel("y"); plt.title("Signal detection: error = shaded overlap")
plt.show()
```

```
P(error) = Phi(-1/sigma) = 0.0228    (sigma = 0.5)
P(error), simulated      = 0.0228
```



2 Joint PDFs

2.1 Joint PDF of continuous random variables

i Definition: Joint PDF

Random variables X, Y are **jointly continuous** with **joint PDF** $f_{X,Y}(x, y)$ if $f_{X,Y}(x, y) \geq 0$ everywhere,

$$\iint_{\mathbb{R}^2} f_{X,Y}(x, y) dx dy = 1,$$

and for every (nice) region $B \subseteq \mathbb{R}^2$,

$$P((X, Y) \in B) = \iint_B f_{X,Y}(x, y) dx dy.$$

i Definition: Marginal PDF

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx.$$

These are the continuous analogs of the discrete cases – sums become integrals!

2.1.1 Example: a 2D uniform density

Let (X, Y) be **uniform on the unit disk** $D = \{(x, y) : x^2 + y^2 \leq 1\}$, i.e.

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{\pi} & (x,y) \in D \\ 0 & \text{otherwise,} \end{cases}$$

since the disk has area π . What is the marginal $f_X(x)$?

$$f_X(x) = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi} dy = \frac{2\sqrt{1-x^2}}{\pi}, \quad -1 \leq x \leq 1.$$

Warning

Even though the **joint** density is perfectly flat (uniform), the **marginal** density of X alone is *not* uniform — it's largest at $x = 0$ (where the disk is widest) and tapers to 0 at $x = \pm 1$. Marginalizing a uniform joint density does not generally give a uniform marginal!

2.1.2 Example: Buffon's needle

A floor is ruled with parallel lines a distance d apart. A needle of length $L \leq d$ is dropped at random. What's the probability it crosses a line?

Setup. Let X = distance from the needle's center to the nearest line, and Θ = acute angle the needle makes with the lines. A "random drop" means

$$X \sim \text{Uniform}(0, d/2), \quad \Theta \sim \text{Uniform}(0, \pi/2), \quad X, \Theta \text{ independent,}$$

so the joint PDF is $f_{X,\Theta}(x, \theta) = \frac{1}{(d/2)} \cdot \frac{1}{(\pi/2)} = \frac{4}{\pi d}$ on the rectangle $[0, d/2] \times [0, \pi/2]$.

The needle **crosses a line** exactly when $X \leq \frac{L}{2} \sin \Theta$ (the needle's half-length, projected perpendicular to the lines, reaches past the center to the line). So

$$P(\text{cross}) = \iint_{\{x \leq \frac{L}{2} \sin \theta\}} \frac{4}{\pi d} dx d\theta = \frac{4}{\pi d} \int_0^{\pi/2} \frac{L}{2} \sin \theta d\theta = \frac{4}{\pi d} \cdot \frac{L}{2} \cdot 1 = \frac{2L}{\pi d}.$$

Note

Solving for π : $\pi = \frac{2L}{d \cdot P(\text{cross})}$ — dropping needles gives a (very slow!) randomized way to estimate π .

```
L, d = 1.0, 2.0
p_cross_formula = 2 * L / (np.pi * d)
print(f"P(cross), formula = {p_cross_formula:.5f}")

n = 2_000_000
X = rng.uniform(0, d / 2, size=n)
Theta = rng.uniform(0, np.pi / 2, size=n)
crosses = X <= (L / 2) * np.sin(Theta)
p_cross_sim = crosses.mean()
```

```
pi_estimate = 2 * L / (d * p_cross_sim)

print(f"P(cross), simulated = {p_cross_sim:.5f}  (n = {n:}, drops)")
print(f"implied estimate of pi = {pi_estimate:.4f}  (true pi = {np.pi:.4f})")
```

```
P(cross), formula = 0.31831
P(cross), simulated = 0.31821  (n = 2,000,000 drops)
implied estimate of pi = 3.1426  (true pi = 3.1416)
```

2.2 Joint CDFs

i Definition: Joint CDF

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u,v) dv du.$$

Same idea as a single-variable CDF: “accumulated probability,” now over a **quadrant** $(-\infty, x] \times (-\infty, y]$ rather than a ray.

2.2.1 Example: uniform on the unit square

Let (X,Y) be uniform on $[0,1]^2$, so $f_{X,Y}(x,y) = 1$ on the square (note: here X,Y are independent Uniform(0,1)’s, unlike the disk example). For $0 \leq x,y \leq 1$,

$$F_{X,Y}(x,y) = \int_0^x \int_0^y 1 dv du = xy.$$

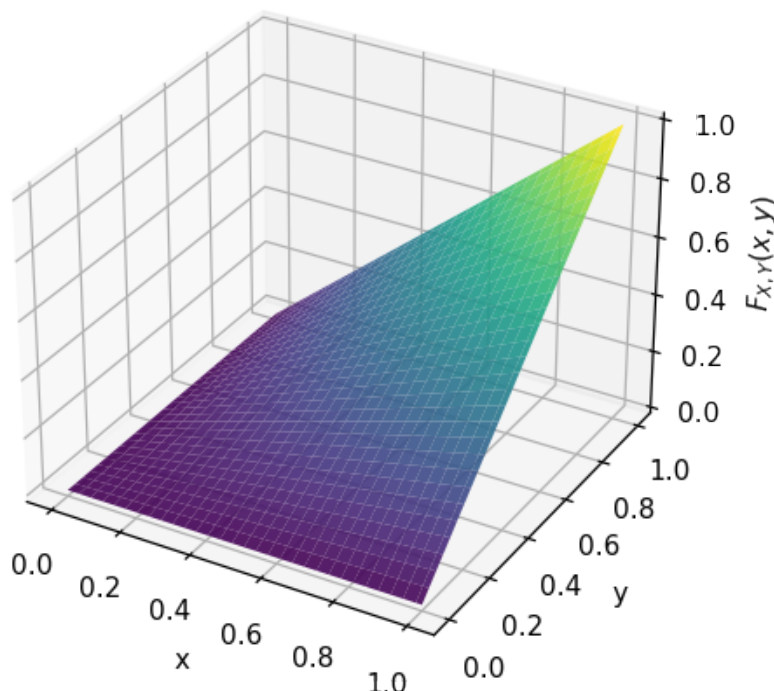
Visual interpretation: since the density is flat (height 1), $F_{X,Y}(x,y)$ is just the probability mass in the rectangle $[0,x] \times [0,y]$, which — because density is 1 — equals the *area* of that rectangle, $x \cdot y$.

```
# Visualize F(x,y) = xy over the unit square, and check it against numerical integration of f=1
xg = np.linspace(0, 1, 60)
yg = np.linspace(0, 1, 60)
Xg, Yg = np.meshgrid(xg, yg)
Fg = Xg * Yg

fig = plt.figure(figsize=(6, 5))
ax = fig.add_subplot(projection="3d")
ax.plot_surface(Xg, Yg, Fg, cmap="viridis", alpha=0.9)
ax.set_xlabel("x"); ax.set_ylabel("y"); ax.set_zlabel(r"$F_{\{X,Y\}}(x,y)$")
ax.set_title("Joint CDF of Uniform on the unit square")
plt.show()

# spot check at one point via double integration of the density
x0, y0 = 0.3, 0.7
F_numeric, _ = dblquad(lambda v, u: 1.0, 0, x0, 0, y0)
print(f"F(0.3, 0.7) via dblquad = {F_numeric:.4f}  (formula x*y = {x0*y0:.4f})")
```

Joint CDF of Uniform on the unit square



$F(0.3, 0.7)$ via `dblquad` = 0.2100 (formula $x*y = 0.2100$)

2.3 Expectation for jointly continuous RVs

i Definition: Two-variable LOTUS

$$E[g(X, Y)] = \iint_{\mathbb{R}^2} g(x, y) f_{X, Y}(x, y) dx dy.$$

This is the continuous counterpart of the two-variable LOTUS for discrete RVs.

💡 Fact: Linearity of expectation

For any jointly continuous X, Y (independent or not!) and constants a, b, c ,

$$E[aX + bY + c] = aE[X] + bE[Y] + c.$$

Again, this needs **no independence assumption** — it follows immediately by taking $g(x, y) = ax + by$ in the two-variable LOTUS and using linearity of the double integral.

2.4 Extending to more than two random variables

Everything above extends directly to a collection $X_1, \dots, X_n$: a joint PDF $f_{X_1, \dots, X_n}$ on $\mathbb{R}^n$ that integrates to 1, marginals obtained by integrating out the variables you don't want, a joint CDF, an n -variable LOTUS, and linearity of expectation for *any* linear combination $\sum_i a_i X_i$ — again with no independence required.

i Definition: covariance

For random variables X_i, X_j ,

$$\text{Cov}(X_i, X_j) := E[(X_i - E[X_i])(X_j - E[X_j])] = E[X_i X_j] - E[X_i]E[X_j].$$

Covariance measures whether X_i and X_j tend to be above/below their means *together* (positive covariance) or in *opposite* directions (negative covariance). Note $\text{Cov}(X_i, X_i) = \text{Var}(X_i)$ — variance is just a variable's covariance with itself.

The covariance matrix. For a random vector $X = (X_1, \dots, X_n)$, Σ is the $n \times n$ matrix whose (i, j) entry is

$$\Sigma_{ij} = \text{Cov}(X_i, X_j).$$

Since $\text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i)$, Σ is symmetric; its diagonal entries $\Sigma_{ii} = \text{Var}(X_i)$ are the individual variances, and its off-diagonal entries capture the pairwise relationships between coordinates.

2.4.1 The Joint PDF of a Multivariate Gaussian

i A first look: multivariate Gaussian random variables

A random vector $(X_1, \dots, X_n)$ is (jointly, or **multivariate**) **Gaussian** if every linear combination $\sum_i a_i X_i$ is a (univariate) normal random variable. We write $X \sim N(\mu, \Sigma)$, where $\mu = (E[X_1], \dots, E[X_n])$ and Σ is the **covariance matrix** — a natural generalization of σ^2 that also records how the X_i vary *together*.

When Σ is invertible, $X \sim N(\mu, \Sigma)$ has joint density

$$f_X(x_1, \dots, x_n) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu)\right),$$

where $x = (x_1, \dots, x_n)$ and $(x - \mu)^\top \Sigma^{-1} (x - \mu)$ is a single number (a quadratic form) — the matrix generalization of $\left(\frac{x - \mu}{\sigma}\right)^2$.

```
# A visual teaser: a 2D Gaussian with correlated components
```

```
rho = 0.6
```

```
mean = [0, 0]
```

```
cov = [[1, rho], [rho, 1]]
```

```
x, y = np.mgrid[-3:3:.05, -3:3:.05]
```

```
pos = np.dstack((x, y))
```

```
rv = stats.multivariate_normal(mean, cov)
```

```
plt.figure(figsize=(5.5, 5))
```

```
plt.contourf(x, y, rv.pdf(pos), levels=20, cmap="viridis")
```

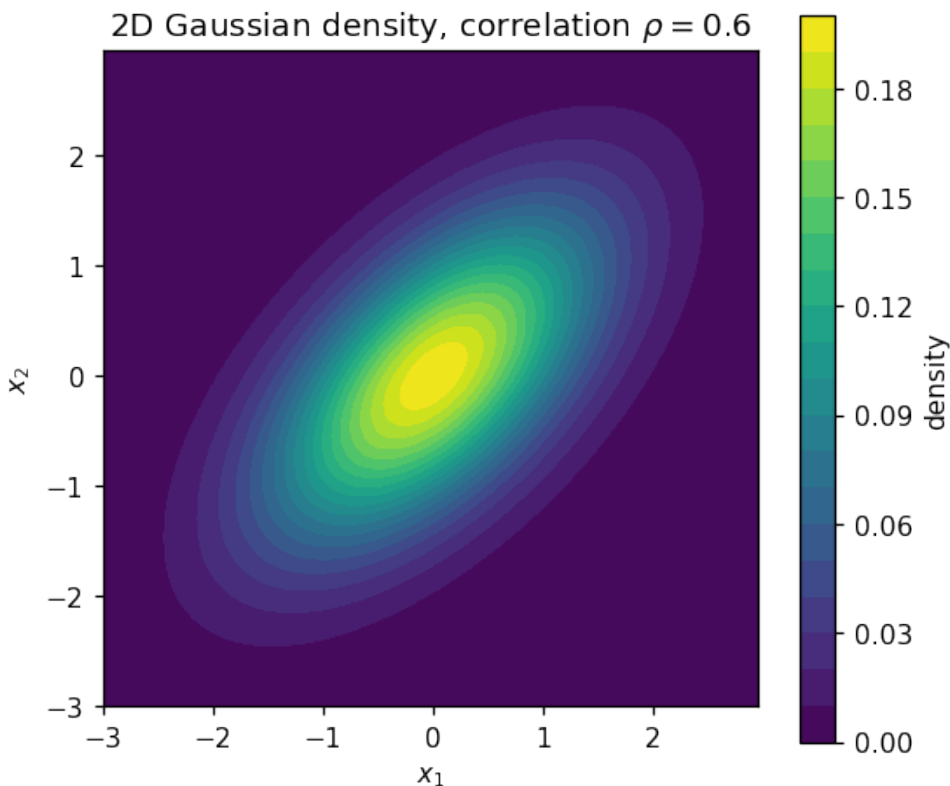
```
plt.gca().set_aspect("equal")
```

```
plt.xlabel(r"$x_1$"); plt.ylabel(r"$x_2$")
```

```
plt.title(fr"2D Gaussian density, correlation $\rho$={rho}")
```

```
plt.colorbar(label="density")
```

```
plt.show()
```



💡 Sanity check: $n = 1$

When $n = 1$, Σ is just the 1×1 matrix $[\sigma^2]$, so $|\Sigma|^{1/2} = \sigma$, $\Sigma^{-1} = 1/\sigma^2$, and the formula collapses to the familiar

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Two facts we'll dig into more soon:

- each individual X_i is itself normal (a marginal of a multivariate Gaussian is Gaussian);
- unlike most joint distributions, the *entire* joint distribution of a multivariate Gaussian depends on only the means and the covariance matrix Σ — visible directly in the formula above.